

day-80-multi-agent-status-consistency

Day 80 - Multi-Agent Status Consistency

Status: PASS for developer-side consistency enforcement

Blocker class: NONE for this lane

Lane

Mission Control, Gateway, Agent Hub, dispatcher, and CloudCode support must agree on the same canonical agent statuses for Agent Zero, Hermes, Pi, Paperclip, SpaceAgent, and OpenClaw+.

What Was Implemented

- Added a canonical multi-agent status consistency report builder.
- Wired '/api/gateway/agent-hub/status' to include 'agent_status_consistency'.
- Reused the CloudCode 'buildAgentHealth' output instead of creating a second health truth source.
- Exported the Agent Hub to CloudCode health id mapper for consistency checks.
- Added regression tests that fail on duplicate agent rows, missing CloudCode health rows, fake active owner status with blockers, and invalid live-interface proof conflicts.
- Preserved truthful partial-runtime states where a Mission Control interface is proven but the backend is blocked, such as Hermes service-down and SpaceAgent credential/service-gated lanes.

Files Changed

- 'src/lib/agent-status-consistency.ts'
- 'src/lib/agent-status-consistency.test.ts'
- 'src/lib/gateway-cloudcode-integration.ts'
- 'src/app/api/gateway/agent-hub/status/route.ts'
- 'runtime/day-80-multi-agent-status-consistency-proof.json'
- 'runtime/day-80-multi-agent-status-consistency.md'
- 'runtime/day-80-multi-agent-status-consistency.pdf'

Routes Changed

- 'GET /api/gateway/agent-hub/status'

The route now returns:

- 'cloudcode_agent_health'
- 'agent_status_consistency'
- canonical registry ids
- Agent Hub ids
- CloudCode health ids
- consistency issues
- execution and write flags fixed to false

UI Behavior

No Gateway designer HTML, CSS, class names, or visual assets were modified.

Agent Hub and Mission Control can now render a single consistency truth:

- no fake LIVE status when a runtime blocker exists
- no duplicate agent status row
- no missing canonical registry row
- no missing CloudCode health row
- no write or execution enabled unless the lane is truly LIVE
- interface proof can coexist with blocked backend truth only for explicit partial-runtime blockers

Service / Runtime Behavior

Runtime proof used a short-lived local-only standalone Mission Control runtime bound to '127.0.0.1:3337'. The proof server was started only for smoke verification and was shut down by the proof script.

No external writes were enabled. No Bridge action was executed. No '.env' file was changed. No API key was printed.

Tests And Checks

- 'pnpm vitest run src/lib/agent-status-consistency.test.ts src/lib/gateway-cloudcode-integration.test.ts src/lib/gateway-agent-hub.test.ts --reporter=dot': PASS, 3 files, 17 tests
- 'pnpm run typecheck': PASS
- 'pnpm run build': PASS
- 'pnpm test': PASS, 200 files, 1490 tests
- 'git diff --check': PASS
- 'node scripts/check-protected-file-invariants.mjs': PASS
- 'node scripts/secret-scan-contract.mjs': PASS
- 'node scripts/raw-exposure-scan-contract.mjs': PASS
- '.env' diff check: clean
- protected route smoke: PASS, 41 routes, 0 failures
- local authenticated Agent Hub consistency smoke: PASS, 6 checked agents, 0 issues

Proof

Proof artifact: 'runtime/day-80-multi-agent-status-consistency-proof.json'

Observed consistency result:

- canonical agents: Paperclip, Agent Zero, Hermes, SpaceAgent, Pi, OpenClaw+
- Agent Hub rows: same 6 agents
- CloudCode health rows: Agent Zero, Hermes, Pi, Paperclip, OpenClaw+, SpaceAgent Playwright, SpaceAgent YouTube, SpaceAgent Firecrawl
- consistency issues: 0
- execution enabled: false
- writes enabled: false
- secrets exposed: false
- raw paths exposed: false

Remaining Blockers

No Day 80 consistency blocker remains.

The following individual agent blockers remain truthful and are carried into later lanes:

- 'agent_zero_external_api_key_missing'
- 'hermes_not_installed'
- 'paperclip_sandbox_service_not_running'
- 'playwright_mcp_service_unreachable'
- 'firecrawl_credential_required'
- 'openclaw_doctor_runtime_not_reachable'

Blocker classification for Day 80: NONE

Rollback

After commit:

```
""bash
git revert <day-80-multi-agent-status-consistency-commit>
""
```

Commit / Push

Pending at report generation time.

Next Day

Day 81 - Multi-Agent Proof Packet automatically starts after this lane closes.